Lecture 7 // Sept 15 // Outline of proof of Donaldson's theorem

Recall
Let (M^4, g) be positive definite and simply connected. Fix a $k = 1$ SU(2) bundle E over M. Set

$$\mathcal{M} := \{\text{SD YM connections on } E\}/\mathcal{G}.$$

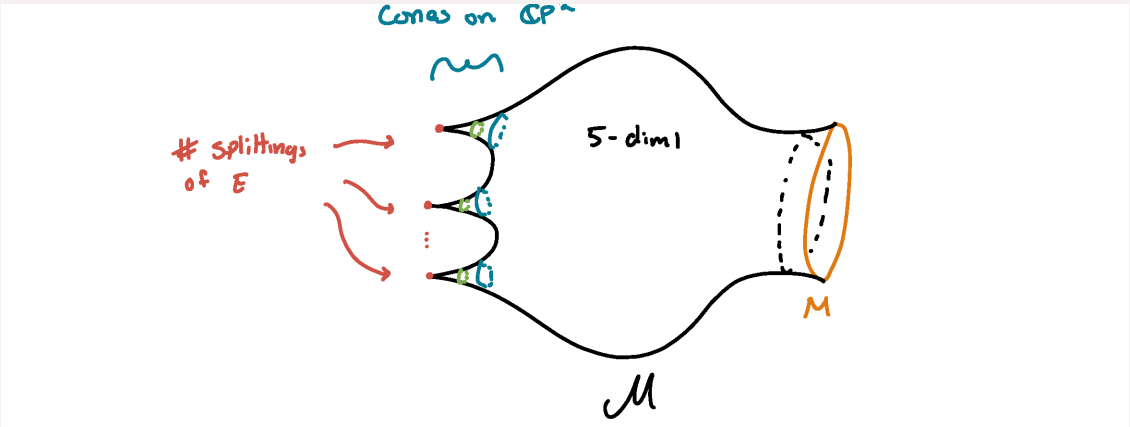
Each splitting $L \oplus \bar{L}$ of E has a unique (up to SU(2) gauge equivalence) split SDYM connection $\nabla \oplus \bar{\nabla}$ for which F_∇ is the unique harmonic representative of $c_1(L)$.

Today, we'll outline a proof of Donaldson's theorem by establishing qualitative features about $\mathcal{M}$.

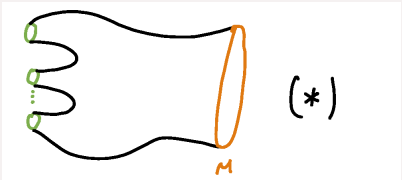
Theorem 1 (Donaldson)
If X^4 is $\pi_1 = 1$, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$.

Claim 2.

1. *Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dimensional oriented manifold.*
2. *The number of these points = the # of splittings of E.*
3. *Each point has a neighborhood homeomorphic to cone on $\mathbb{C}\mathrm{P}^2$; diffeomorphic away from point.*
4. *$\mathcal{M}$ is <u>not</u> compact, but it admits a compactification $\bar{\mathcal{M}}$ such that $\partial\bar{\mathcal{M}} \cong M$.*



We have a cobordism M to $\bigsqcup_{\substack{\#\text{splittings} \\ \text{of } E}} \mathbb{C}\mathrm{P}^2$ obtained by removing the singular points. $(*)$

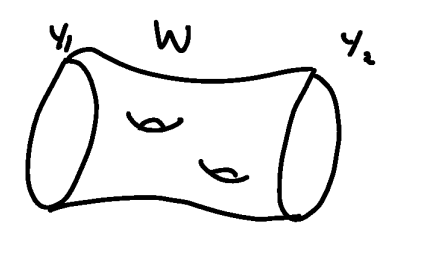


It turns out that $(*)$ implies Donaldson's theorem. Before we discuss this, we'll briefly motivate why bordism is useful here.

Interlude: Bordism in General

This discussion isn't strictly necessary, but it places $(*)$ in context.

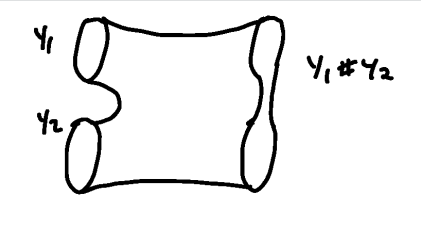
Definition 3



The n-th bordism group is

$$\Omega^n = \{\text{Set of all oriented smooth } n\text{-mfd.}\} / \sim$$

where $Y_1 \sim Y_2$ iff there exists W^{n+1} such that $\partial W = -Y_1 \sqcup Y_2$. If $Y_1 \sim Y_2$, they are called bordant. Ω^n is a group under disjoint union and/or connect sum.



This is a generalization of the question: is $M^n \cong \partial W^{n+1}$ for some W^{n+1}? It turns out that in one of the great algebraic topology triumphs of the 60's, the groups Ω^n are well understood. In particular, Ω^n/torsion is known for all $n \in \mathbb{N}$. Concretely, for $n = 3$, $\Omega^3 = 0$ since every 3-manifold bounds a 4-manifold. When $n = 4$, $\Omega^4 \cong \mathbb{Z}$, which is determined by signature.

Definition 4

The signature of a 4-manifold is $\sigma(M) = \sigma(Q_M) = \#+\text{eigenvalues} - \#\text{-eigenvalues}$.

Theorem 5

In dimension 4, signature is a complete bordism invariant.

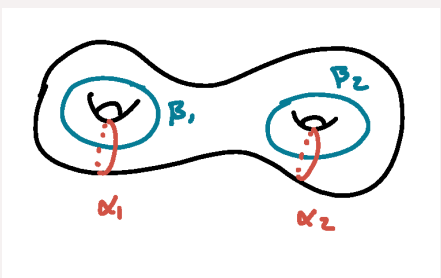
This means that two 4-manifolds are cobordant if and only if they have the same signature. In particular, any M is cobordant to either $\#_{\sigma(M)}\mathbb{C}\mathrm{P}^2$, if $\sigma(M) \geq 0$, or $\#_{-\sigma(M)}\overline{\mathbb{C}\mathrm{P}^2}$, if $\sigma(M) < 0$.

For our purposes, we'll only need the claim that σ is a bordism invariant.

Proposition 6

If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

To prove this proposition, we'll need the following lemma, commonly referred to as the "half lives, half dies" theorem.



To motivate it, consider $Z_g = \partial H_g$. Think of Z_g as a stand in for M and H_g a stand in for W. The middle dimension of the

(co)homology of Z is H_1, so we look at the map induced by the inclusion $\iota : Z_g \to H_g$.

$$\iota_* : H_1(Z_g) \longrightarrow H_1(H_g)$$

$$\ker \iota_* \cong \text{span}\{\alpha_1, \alpha_2\}$$

$$\text{im}\, \iota_* \cong \text{span}\{\iota_*(\beta_1), \iota_*(\beta_2)\}$$

Lemma 7 (Half lives, half dies)
For W a smooth oriented $2n+1$ dimensional manifold with boundary $\partial W \cong M$, $\ker(\iota_*) \subset H_n(M;\mathbb{R})$ is a half-dimensional Lagrangian subspace with respect to the intersection form $Q : H^n(M;\mathbb{R}) \times H^n(M;\mathbb{R}) \to \mathbb{Z}$, where $\iota_* : H_n(M;\mathbb{R}) \to H_n(W;\mathbb{R})$ is the map on homology induced by the inclusion $\iota : M \cong \partial W \hookrightarrow W$.

Proof of Lemma. First recall that by Poincaré duality, $H_n(M;\mathbb{R}) \cong H^n(M;\mathbb{R})$.

Further, from linear algebra, if $T : A \to B$ is a linear map, the dual map $T^* : B^* \to A^*$ is such that $\dim \text{im}\, T = \dim \text{im}\, T^*$:

$$\dim \text{im}\, T = \dim(A) - \dim(\ker T) = \dim(A/\ker T) = \dim(\text{Ann}(\ker T)) = \dim \text{im}\, T^*$$

$$(\text{Ann}(\ker T) \text{ is the subspace of } A^* \text{ that annihilates } T)$$

Thus the maps

$$\iota^* : H^n(W;\mathbb{R}) \longrightarrow H^n(M;\mathbb{R}), \qquad \iota_* : H_n(M;\mathbb{R}) \longrightarrow H_n(W;\mathbb{R})$$

are dual in the linear algebra sense, so $\dim \text{im}\, \iota^* = \dim \text{im}\, \iota_*$.

Again by Poincaré duality, $\dim \text{im}\, \iota^* = \dim \ker \iota_*$.

Thus

$$\dim \ker \iota_* + \dim \text{im}\, \iota_* = \dim H_n(M;\mathbb{R})$$

$$\dim \text{im}\, \iota^* + \dim \text{im}\, \iota^*$$

$$\Rightarrow \dim \text{im}\, \iota^* = \tfrac{1}{2} \dim H_n(M;\mathbb{R}) = \tfrac{1}{2} \dim H^n(M;\mathbb{R})$$

$$\Rightarrow \dim \ker \iota_* = \tfrac{1}{2} \dim H_n(M;\mathbb{R})$$

Now, take $\alpha, \beta \in H^n(W;\mathbb{R})$, so $\iota^*(\alpha), \iota^*(\beta) \in H^n(M;\mathbb{R})$.

Then

$$Q_M\big(\iota^*(\alpha), \iota^*(\beta)\big) = \int_M \iota^*(\alpha) \wedge \iota^*(\beta) = \int_M \iota^*(\alpha \wedge \beta) = \int_{\partial W} \iota_*(\alpha \wedge \beta)$$

By Stokes' theorem,

$$\int_{\partial W} \iota_*(\alpha \wedge \beta) = \int_W d(\alpha \wedge \beta) = 0.$$

Thus $\text{im}(\iota^*)$ is Lagrangian with respect to Q_M. □

Remark 8
We may avoid using Stokes' theorem by translating to algebraic topology and cupping against the fundamental class. Thus half lives half dies is true when M isn't smoothable.

Having proven half lives half dies, we may now prove the original proposition.

Proposition 9
If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

Proof.

1. By setting $M = -M_1 \# M_2$, we know that $M \cong \partial(W^5)$. So it is enough to show that $\sigma(M) = 0$.

2. Consider the long exact sequence in homology and the Poincaré-Lefschetz isomorphism. By naturality, the following diagram commutes:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & H^2(W;\mathbb{R}) & \xrightarrow{\iota^*} & H^2(M;\mathbb{R}) & \longrightarrow & H^3(W,M;\mathbb{R}) & \longrightarrow & \cdots \\ & & \Big\downarrow{\scriptstyle PD}\cong & & \Big\downarrow{\scriptstyle PD}\cong & & \Big\downarrow{\scriptstyle PD}\cong & & \\ \cdots & \longrightarrow & H_3(W,M;\mathbb{R}) & \longrightarrow & H_2(M;\mathbb{R}) & \xrightarrow{\iota_*} & H_2(W;\mathbb{R}) & \longrightarrow & \cdots \end{array}$$

   Note that $\operatorname{im}\iota^* \cong \ker \iota_*$ under PD.

3. By Half lives, half dies,

$$\begin{cases} \dim \operatorname{im}\iota^* = \frac{1}{2}\dim H^2(M;\mathbb{R}) \\ \dim \ker \iota_* = \frac{1}{2}\dim H_2(M;\mathbb{R}) \end{cases}$$

   and Q_M vanishes on $\operatorname{im}\iota^*$ (e.g., $\operatorname{im}\iota^* \subset H^2(M;\mathbb{R})$ is Lagrangian).

   Thus there is a half-dimensional subspace on which the intersection form vanishes.

4. This implies $\sigma(M) = 0$: consider the diagonalization of Q_M over $\mathbb{R}$: $Q_M \cong \begin{pmatrix} +I & 0 \\ 0 & -I \end{pmatrix}$.

$$\begin{pmatrix} +I & 0 \\ 0 & -I \end{pmatrix} V = V_+ \oplus V_-$$

   This splits V into $V_+ \oplus V_-$, where Q_M is $\pm$-definite on $V_\pm$.

   Let $a = \dim V_+$ and $b = \dim V_-$. Then $a + b = \dim V_+ + \dim V_- = \dim V = 2k$.

   Let $U = \operatorname{im}\iota^*$. Since Q_M vanishes on U, $U \cap V_+ = \{0\}$ and $U \cap V_- = \{0\}$.

   Thus

$$\dim(U + V_+) = \dim(U) + \dim(V_+) = k + a \leq 2k$$

$$\dim(U + V_-) = \dim(U) + \dim(V_-) = k + b \leq 2k$$

$$\Rightarrow a \leq k,\ b \leq k. \text{ Since } a + b = 2k,\ a = b = k \Rightarrow \sigma(M) = a - b = 0. \qquad \square$$

Back to Donaldson

Let's return to Donaldson and take stock of what we know:

Theorem 10 (Donaldson)
If X^4 is $\pi_1 = 1$, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$.

The previous discussion of bordism has given us the following three facts, from which Donaldson's theorem will follow.

(i) $\sigma(M) = \mathrm{rk}(Q_M)$ (since M is positive definite).

(ii) $\sigma(M) = \#$ splittings of E.

Since

(a) σ bordism invariant (Interlude)

(b) M bordant to $\bigsqcup_{\substack{\#\text{splittings} \\ \text{of } E}} \mathbb{C}\mathrm{P}^2$ $(*)$

(iii) $\#$ splittings of $E = \frac{1}{2}\#x \in H^2$ such that $\langle x, x\rangle = 1$.

This follows from two claims about vector bundles over X^4:

(a) SU(2) bundles are determined by $c_2 \in H^4$ (true only in our setting)

(b) U(1) bundles are determined by $c_1 \in H^2$ (true in general)

Via the curvature form or by formal properties of Chern classes, to say $E = L \oplus \bar{L}$ implies that $c_2(E) = c_1(L) \cup c_1(\bar{L}) = -c_1(L) \cup c_1(L)$.

Similarly, if $c_2(E) = -x^2$ for $x \in H^2$, the above calculation implies that $E = L \oplus \bar{L}$. Note here that different choices of x give different L's in the splitting.

Therefore, if $k = c_2(E) = 1$, we must search over $x \in H^2$ for which $x^2 = 1$.

Combining (i), (ii), and (iii), we can see that

$$\sigma(M) = \mathrm{rk}(Q_M) = \#\text{splittings of } E = \tfrac{1}{2}\#x \in H^2 \text{ s.t. } x^2 = 1.$$

Heuristically, this equality says that Q_M has "many" x with $\langle x, x\rangle = 1$, where "many" is exactly twice the number of splittings.

To prove Donaldson's theorem, it remains to show that Q_M is equivalent to $n(+1)$ or $n(-1)$, i.e., $\pm n = \mathrm{rk}(Q_M) = \frac{1}{2}\#x \in H^2$ such that $x^2 = 1$.

Lemma 11
Let $Q : V \times V \to \mathbb{Z}$ be a positive-definite symmetric unimodular bilinear pairing. Then $\#x$ such that $Q(x, x) = 1 \leq 2\,\mathrm{rk}\,Q$, with equality iff Q is equivalent to I over $\mathbb{Z}$.

Assuming the lemma, since $\mathrm{rk}(Q_M) = \frac{1}{2}\#x \in H^2$ such that $x^2 = 1$, where x^2 is shorthand for $Q(x,x)$, we get that

$$\#x \text{ s.t. } Q(x,x) = 1 \leq 2 \cdot \tfrac{1}{2}\#x \in H^2 \text{ s.t. } Q(x,x) = 1$$

$$\#x \text{ s.t. } Q(x,x) = 1 \leq \#x \in H^2 \text{ s.t. } Q(x,x) = 1$$

So clearly we must have equality, so the lemma implies that Q is equivalent to I over $\mathbb{Z}$, thus $Q \cong \pm(n)$. This proves Donaldson's theorem.

We now prove the lemma.

Proof of Lemma. First note that if $Q = I$, there is clearly $2 \cdot \mathrm{rk}\, Q$ of such elements of H^2, since the standard basis vectors e_i have self pairing 1, as well as $-e_i$.

For a general Q, we'll apply the Gram-Schmidt process over $\mathbb{Z}$. Let x be such that $Q(x,x) = 1$. We claim that $V = \langle x \rangle \oplus \langle x \rangle^\perp$. Indeed, since any $z \in V$ decomposes as

$$z = \underbrace{Q(x,x)x}_{\in \langle x \rangle} + \underbrace{(z - Q(x,z)x)}_{\in \langle x \rangle^\perp}$$

we know that $V = \langle x \rangle + \langle x \rangle^\perp$; since $\langle x \rangle \cap \langle x \rangle^\perp = \{0\}$, this decomposition is orthogonal.

Thus $Q : \langle x \rangle \oplus \langle x \rangle^\perp \times \langle x \rangle \oplus \langle x \rangle^\perp \to \mathbb{Z}$, so Q splits as a block matrix

$$Q = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & * \end{array}\right) \begin{array}{l} \left.\vphantom{\begin{array}{c}1\\0\end{array}}\right\} \langle x \rangle \\ \left.\vphantom{\begin{array}{c}1\\0\end{array}}\right\} \langle x \rangle^\perp \end{array}$$

Now, notice the following:

1. Q is still positive definite on $x^\perp$.
2. Any $z \in V$ such that $Q(z,z) = 1$ is either $\pm x$ or lies entirely in $\langle x \rangle^\perp$: if $z = cx + w$ for $w \in x^\perp$, $Q(z,z) = c^2 + \langle w, w \rangle \neq 1$.
3. The rank of $Q|_{\langle x \rangle^\perp}$ is $\mathrm{rank}(Q) - 1$.

Thus we may proceed by induction on $Q|_{\langle x \rangle^\perp} : \langle x \rangle^\perp \times \langle x \rangle^\perp \to \mathbb{Z}$. If $\langle x \rangle^\perp$ contains no elements for which $x^2 = 1$, the process concludes, and the number of elements for which $x^2 = 1$ is necessarily less than $2 \cdot \mathrm{rk}(Q)$. Otherwise, Q can be completely diagonalized, so Q is equivalent to I. □